

PROJECT PROPOSAL

Cryptocurrency Historical Price Analysis and Visualization Using Python

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Project Type	Individual Project
Dataset Source	Kaggle – Cryptocurrency Historical Prices

1. Project Definition

The proposed project aims to analyze historical cryptocurrency price data using Python. The project will study cryptocurrency price movements, trading volume, and market-related trends over time. Data analysis and visualization techniques will be used to identify patterns, price fluctuations, and differences in the performance of selected cryptocurrencies.

2. Dataset Use Case

The project will use a cryptocurrency historical-price dataset obtained from Kaggle. The dataset will be used for exploratory data analysis and time-series analysis. Historical price information will be cleaned and analyzed to understand how cryptocurrency prices and trading activity change over time. Selected cryptocurrencies can be compared using their historical closing prices, trading volume, and other available numerical attributes.

Dataset source: <https://www.kaggle.com/search?q=Cryptocurrency%20Historical%20Prices>

3. Visualization Graphs / Plots and Expected Outcomes

Visualization	Expected Outcome
Closing Price vs Date – Line Graph	Shows historical price trends and major increases or decreases over time.
Trading Volume vs Date – Line Graph	Shows changes in trading activity and periods of higher or lower market participation.
Average Price Comparison – Bar Graph	Compares the average price of selected cryptocurrencies.
Volume vs Closing Price – Scatter Plot	Helps examine whether changes in trading volume are associated with changes in closing price.
Market Capitalization Trend – Line Graph	Shows how the market value changes over the selected period, if market-cap data is available.

4. Python Libraries and Tools

Python: Main programming language for the project.

Pandas: Data loading, cleaning, filtering, and analysis.

NumPy: Numerical calculations and data processing.

Matplotlib: Creating line, bar, and other plots.

Seaborn: Statistical and attractive data visualizations.

Jupyter Notebook / Google Colab: Environment for developing and presenting the analysis.

GitHub: Version control, project code, graphs, and documentation.

5. GitHub Repository and Documentation

A GitHub repository will be created for the project. The repository will contain the project documentation (README.md), Python/Jupyter Notebook code, visualization outputs, and the dataset or dataset source/reference as appropriate. Since this is an individual project, the repository will contain the student's own commits.

GitHub Repository: To be added after repository creation.

6. Expected Final Outcome

The final project is expected to provide a clear visual and analytical understanding of historical cryptocurrency market behavior. The analysis will highlight price trends, trading-volume changes, comparisons between selected cryptocurrencies, and relationships among important numerical variables.